

TABLE OF CONTENTS

Table of Contents	
1	COVER PAGE
2	POWER DELIVERY
3	I.MXRT1064_GPIO
4	CONNECTORS_COMMUNICATION
5	ETHERNET
6	IMU
7	LED

REVISION HISTORY

REVISION	DESCRIPTION OF CHANGE	DATE	Approved
A	Release of Final Schematic	10/22/2024	Iain Galloway
A1	Changed BT1 to DNP, Li-Ion batteries cannot be shipped with boards.	10/22/2025	Iain Galloway
B	Completely removed BT1 from schematic. Li-Ion batteries soldered or loose cannot be shipped with NXP boards.	10/22/2025	Iain Galloway
C	Several changes made to schematic per document PCML-94468_C.xlsx. New USB Connector J1 added. USB Conn changed to Full SMT.	1/22/2025	Iain Galloway
D	J1 Connector made SBU1 SBU2 NC. Updated footprint on U13. Added 47uF to input U13	6/4/2025	Iain Galloway
E	Removed GND from pin 3 on J13, J12	9/26/2025	Iain Galloway
F	Changing Net BMI088_DRDY2 from INT2 (pin 1) to INT3 (Pin 12) on U9	7/23/2026	Iain Galloway

MR-VMU-TROPIC

SILKSCREEN

Silkscreen text is this COLOR in the schematic.

CAD NOTES:

CAD Notes are this COLOR in the schematic.

		Mobile Robotics Group 6501 William Cannon Drive West Austin, TX 78735/8598	
ICA Classification: Public - Open Source			
Designer: Michael Steffen	Drawing Title: MR-VMU-TROPIC		
Drawn By: Michael Steffen	Page Title: 1 - COVER PAGE		
Approved: Iain Galloway	Size: C	Document Number: SCH-94468 PDF: SPF-94468	Rev: F
Date: July 23, 2026	Sheet: 1	of 7	

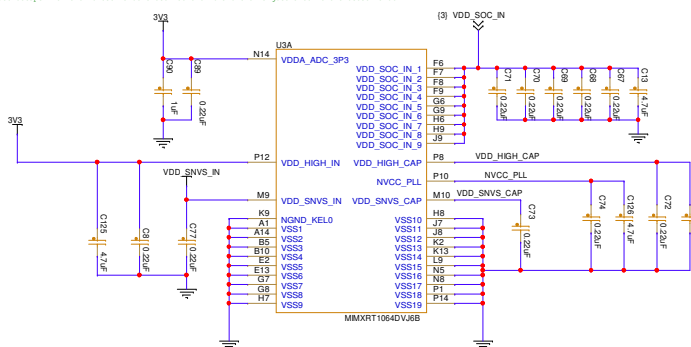
CAD: Please make sure there is adequate thermal relief for this USB connector and U13.

[illegible]

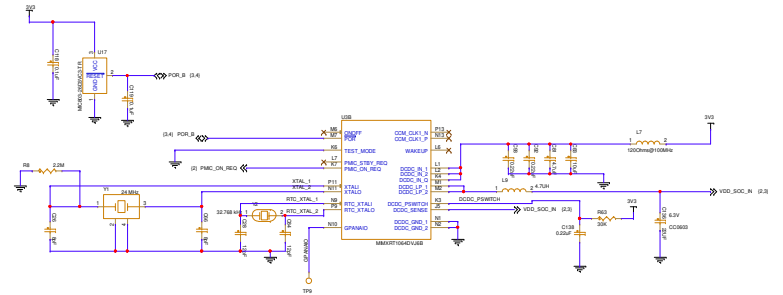
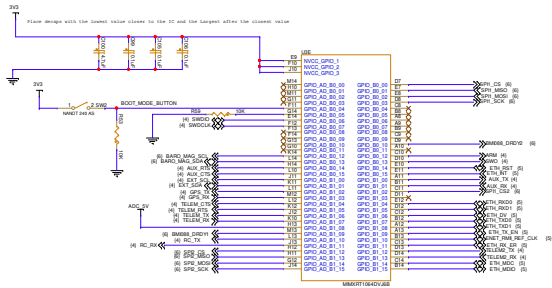
SE BATTERY BACKUP
SOLDER PADS LOCATED ON PCB

Place 0.22uF next to F6 & F7, 0.22uF next to F8 & F9 and so on.

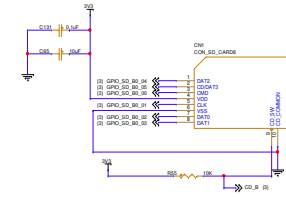
Place decaps with the lowest value closer to the IC and the Largest after the closest value



	Mobile Robotics Group 65.01 W William Cannon Drive West Austin, TX 78735/8596		
ICA Classification: Public - Open Source			
Designer: Michael Steffen	Drawing Title: MR-VMU-TROPIC		
Drawn By: Michael Steffen	Page Title: 2 - POWER DELIVERY		
App. lead: Gary Galloway	Size of Date: July 23, 2006	Document Number: SCH-94468, PDF: SPF-94468 Sheet: 2 of 3	Rev 0

[illegible]

The schematic diagram shows a 12-bit DAC circuit. It includes a 1.2V supply, a 10k resistor network, and two op-amp buffers (U1, U2). The DAC output is connected to a 10k resistor network and a 10k resistor. The output is also connected to a 10k resistor network and a 10k resistor. The output is also connected to a 10k resistor network and a 10k resistor.



		Mobile Robotics Group 6501 Wilshire Center Drive West Suite 710 735/8598	
ICA Classification: Public / Open Source			
Designer: Michi Saffer		Drawing Title: MR_VMR10-ROIC	
Design By: Michi Saffer		Page Title: 3 - LMXRT1064_G PK	
App approved: John Roderick		Size 10 x 15 x 20 mm	Document Number: SCH48465 PDF: SCH48465
		Rev: 1.0	

[illegible]

PWM

(3) PWM_1

(3) PWM_2

(3) PWM_3

RN1

1 2 3 4

8 7 6 5

220

(3.4) ONE_WIRE

J19

1 2 3 4 5 6 7 8

PLUG_1X8

PWM_A

TELEM 1 & 2

The diagrams show the internal wiring for two telemetry channels, TELEM 1 and TELEM 2. Both channels use a 5V supply and a ground connection. TELEM 1 uses an NFM18PC104R1C30 IC (labeled FL3) and a resistor network RN3. TELEM 2 uses an NFM18PC104R1C30 IC (labeled FL4) and a resistor network RN4. The signal lines for each channel are connected to a COM1X6 connector, which is labeled with pin numbers 1 through 8. The connections for TELEM 1 are: (3) TELEM TX to pin 1, (3) TELEM RX to pin 2, (3) TELEM CTS to pin 3, and (3) TELEM RTS to pin 4. The connections for TELEM 2 are: (3) TELEM2 TX to pin 1, (3) TELEM2 RX to pin 2, (3) TELEM2 CTS to pin 3, and (3) TELEM2 RTS to pin 4. The COM1X6 connector is also labeled with pin numbers 1 through 8 and a label COM1X6.

[illegible]

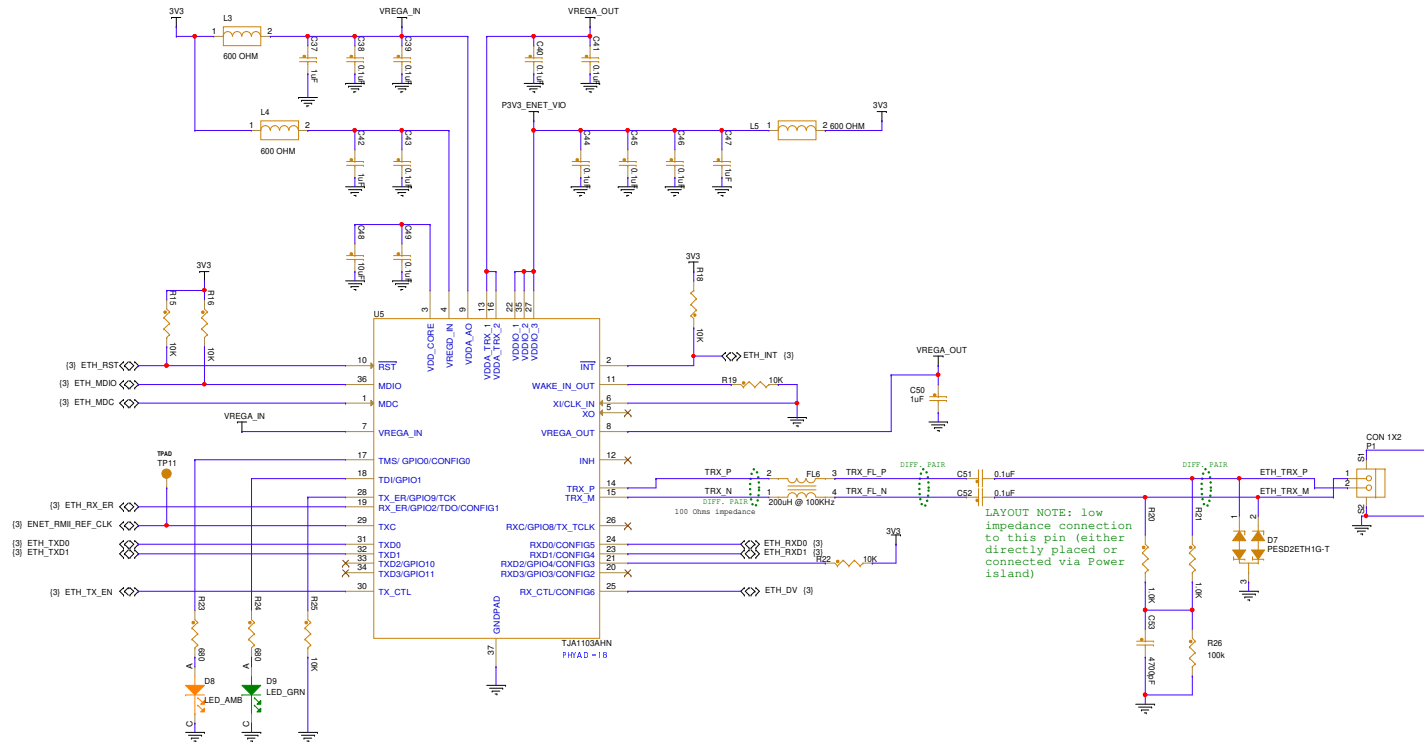
CAN & CAN TRANSCEIVER

[illegible]

RC-ELRS

		Mobile Robotics Group 6501.1 William Cannon Drive West Austin, TX 78753/8598	
ICA Classification: Public - Open Source			
Designer: Michael Steffen		Drawing Title: MR-VMU-TROPIC	
Drawn By: Michael Steffen		Page Title: 4 - CONNECTORS_COMMUNICATION	
Approved: Iain Galavay		Site Code	Document Number: SCH-94468 PDF: SPF-94468
		Date: July 23, 2026	Sheet 4 of 7
			Rev F

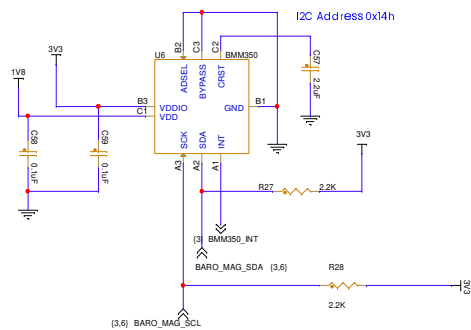
ETHERNET 100BASE-T1



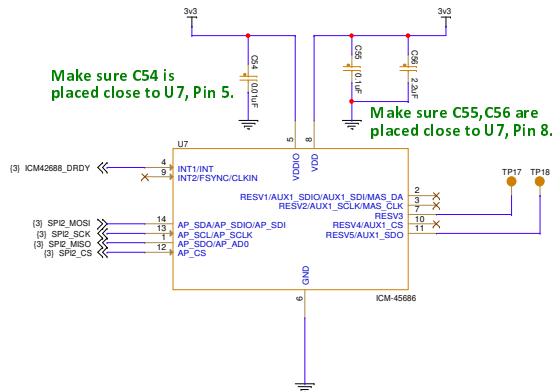
Move D8, D9 LED to top side of PCB.

		Mobile Robotics Group 6501 William Cannon Drive West Austin, TX 78735/8598	
ICA Classification: Public - Open Source			
Designer: MichaelSteffen	Drawing Title: MR-VMU-TROPIC		
Drawn By: MichaelSteffen	Page Title: 5 - ETHERNET		
Approved: Jatin Galloway	Size: C	Document Number: SCH-944.68 PDF: SPF-944.68	Rev: F
	Date: July 23, 2026	Sheet: 5 of 7	

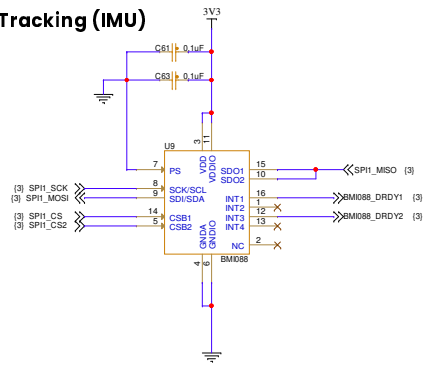
MAGNETOMETER



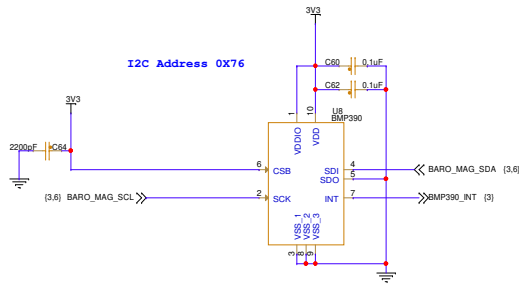
Accelerometer, Gyroscope, 6-Axis Sensor Output



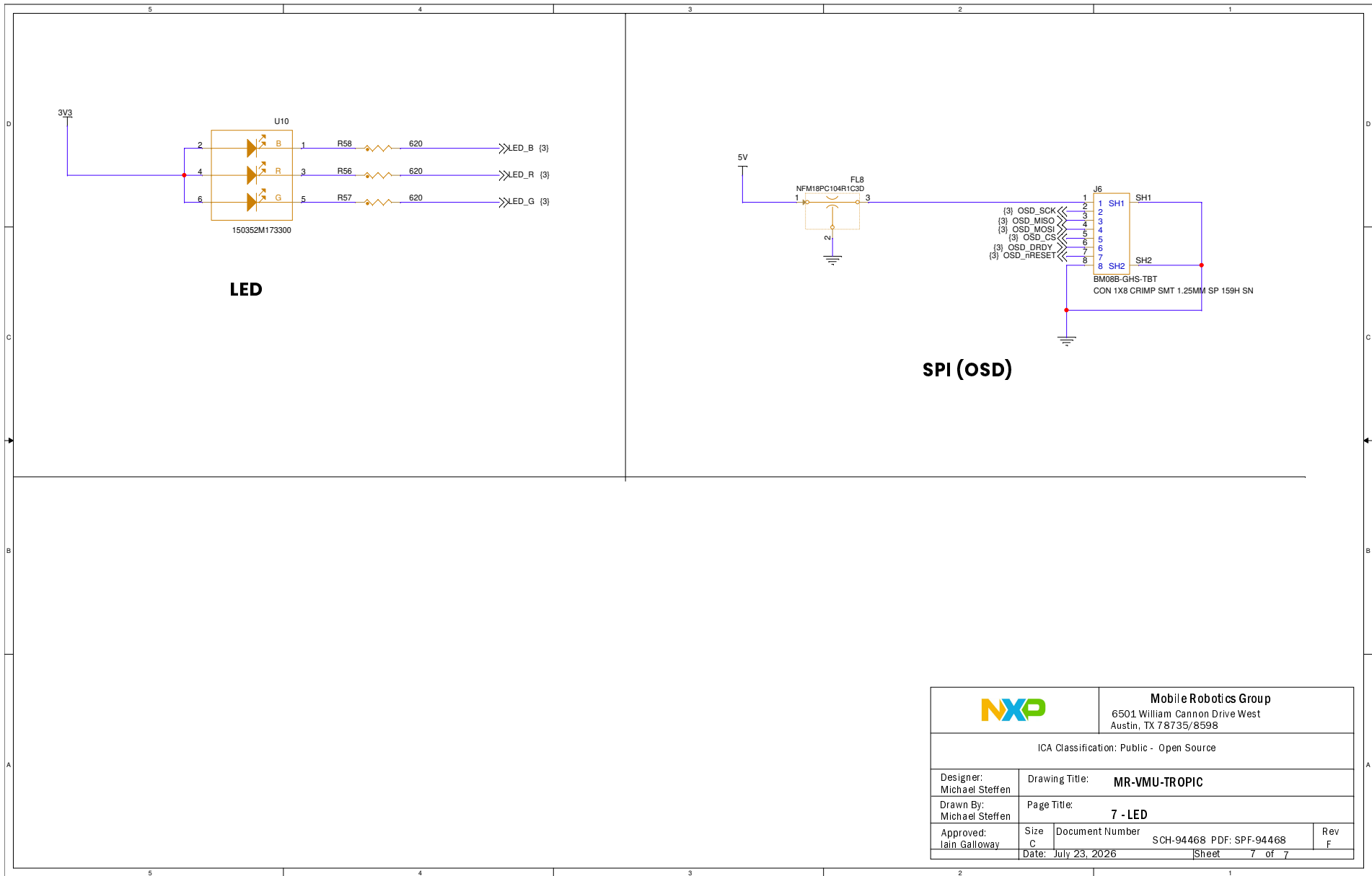
6-Axis Motion Tracking (IMU)
Gyroscope



Pressure Sensor



		Mobile Robotics Group	
		6501 William Cannon Drive West Austin, TX 78735/8598	
ICA Classification: Public - OpenSource			
Designer: MichaelSteffen	Drawing Title: MR-VMU-TROPIC		
Drawn By: MichaelSteffen	Page Title: 6 - IMU		
Approved: Iain Galloway	Size C	Document Number SCH-94468 PDF: SPF-94468	Rev F
Date Jul 23, 2026	Sheet 6 of 7		



		Mobile Robotics Group 6501 William Cannon Drive West Austin, TX 78735/8598		
ICA Classification: Public - Open Source				
Designer: Michael Steffen	Drawing Title: MR-VMU-TROPIC			
Drawn By: Michael Steffen	Page Title: 7 - LED			
Approved: Iain Galloway	Size C	Document Number SCH-94468 PDF: SPF-94468	Rev F	
Date: July 23, 2026		Sheet 7 of 7		